# IMPROVEMENT & ENHANCEMENT

VAULT™

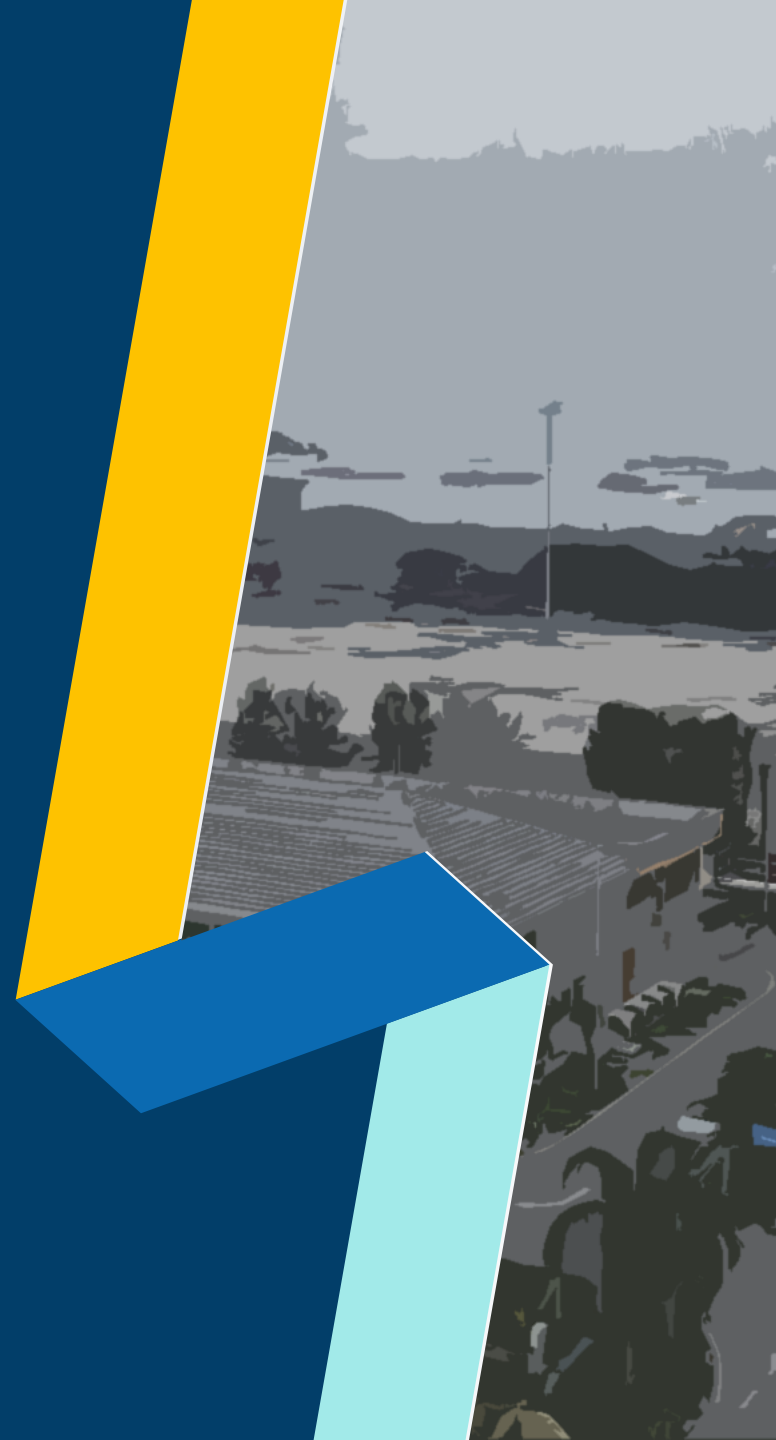
A Health, Safety & Environment (HSE) management solutions invented for Lumut Maritime Terminal Sdn Bhd

Date : 12 May 2022

# ITEM TO IMPROVE ENHANCE

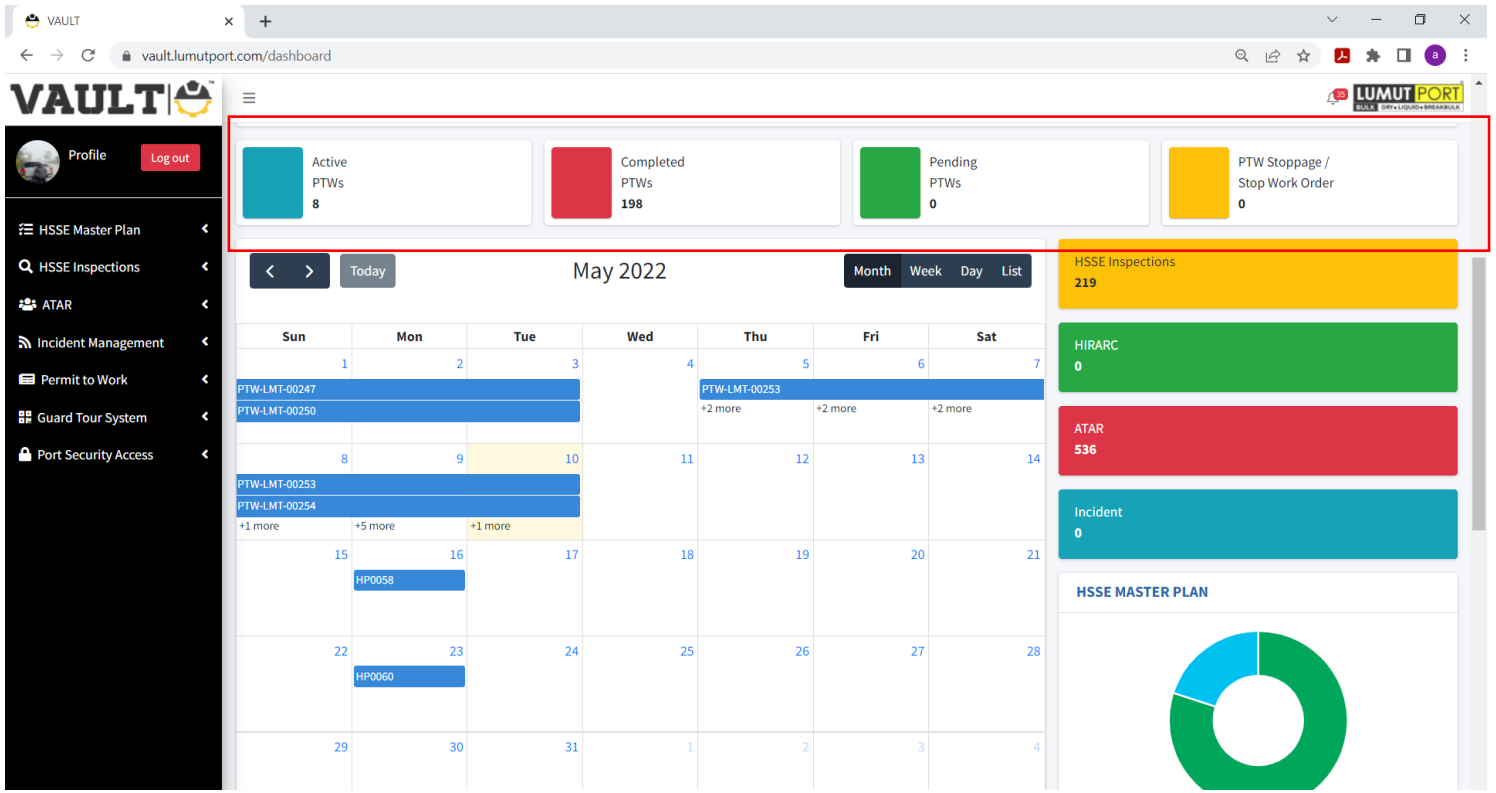
| | |
|---|---|
| **ITEM 1 :** | PERMIT TO WORK (PTW) MODULE |
| **ITEM 2 :** | INCIDENT MANAGEMENT MODULE |
| **ITEM 3 :** | MONTHLY INSPECTION MODULE |
| **ITEM 4:** | ADDITIONAL MODULE FOR ENVIRONMENTAL MONITORING |
| **ITEM 5 :** | GUARD TOUR SYSTEM (GTS) |

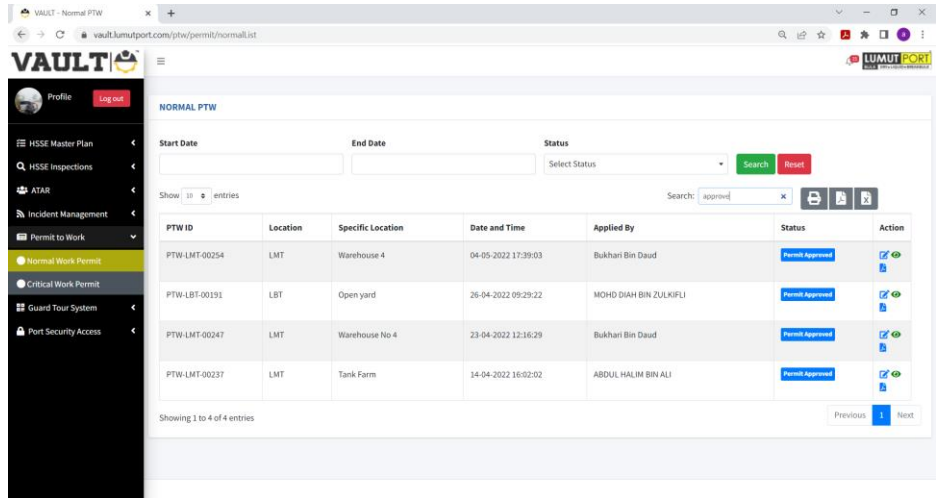
# ITEM 1
## PERMIT TO WORK (PTW)
## MODULE



# ITEM 1.1 PERMIT TO WORK (PTW) MODULE

- Module: PERMIT TO WORK (PTW)
- Problem/Issue to Improve: All PTW status (Active, Complete, etc) showed in dashboard shall have function to select and directly display the list of PTW status (Active, Complete, etc). Accordingly. (Propose to have Dropdown function)



# ITEM 1.2 PERMIT TO WORK (PTW) MODULE

- Module: PERMIT TO WORK (PTW)
- Problem/Issue to Improve: To have additional declaration on PTW Sanction Form. The declaration note/statement will be provided.

LUMUT PORT
BULK DRY • LIQUID • BREAKBULK

**PERMIT TO WORK**

PTW-LMT-00254

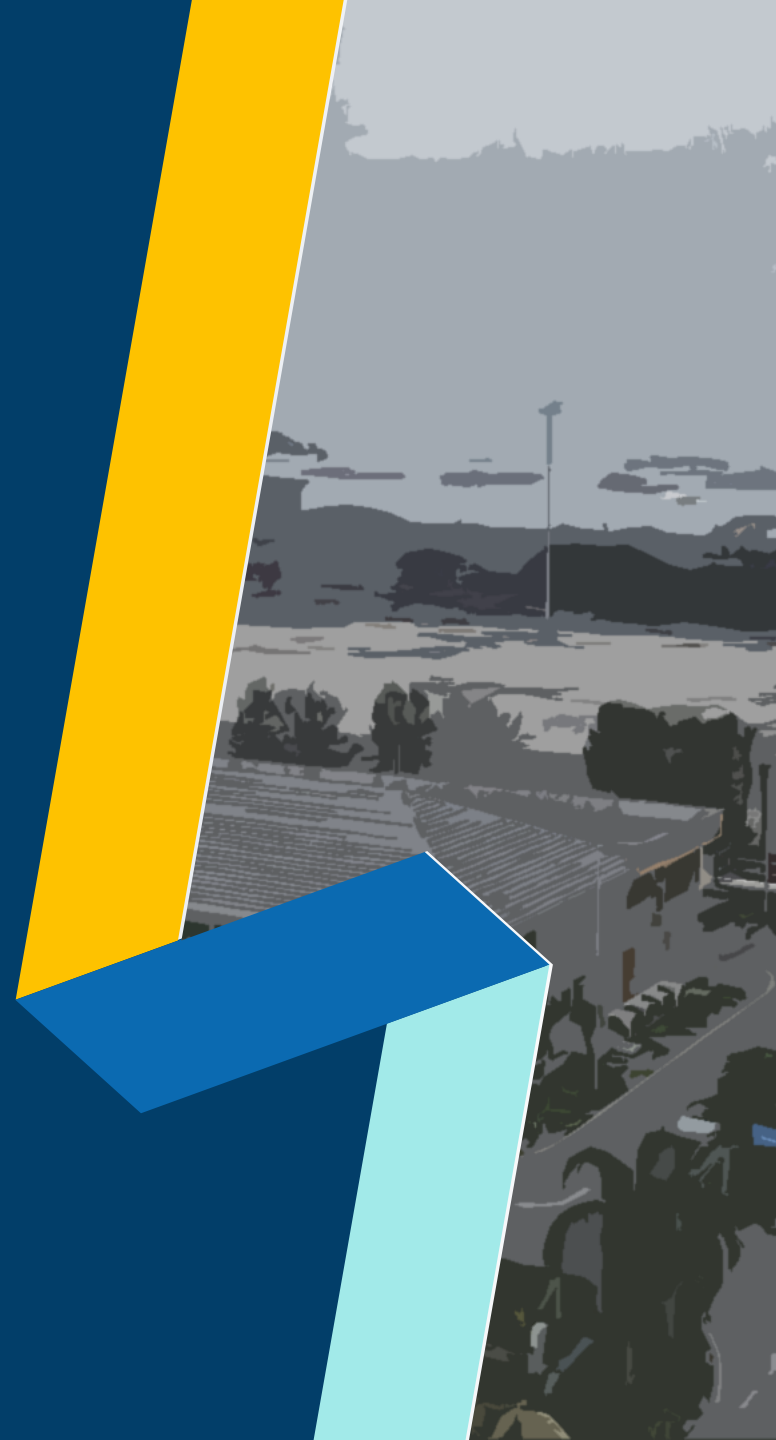
| Date/Time (24hrs) | Work Supervisor Name | Signature |
|---|---|---|
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |

**SECTION 5 - JOINT VISIT BEFORE WORK COMMENCES**

☑ I understand the required Checkbox precautions specified in Section 3 in order to perform activity stipulated on section 2 and accept all responsibilities for the work stated

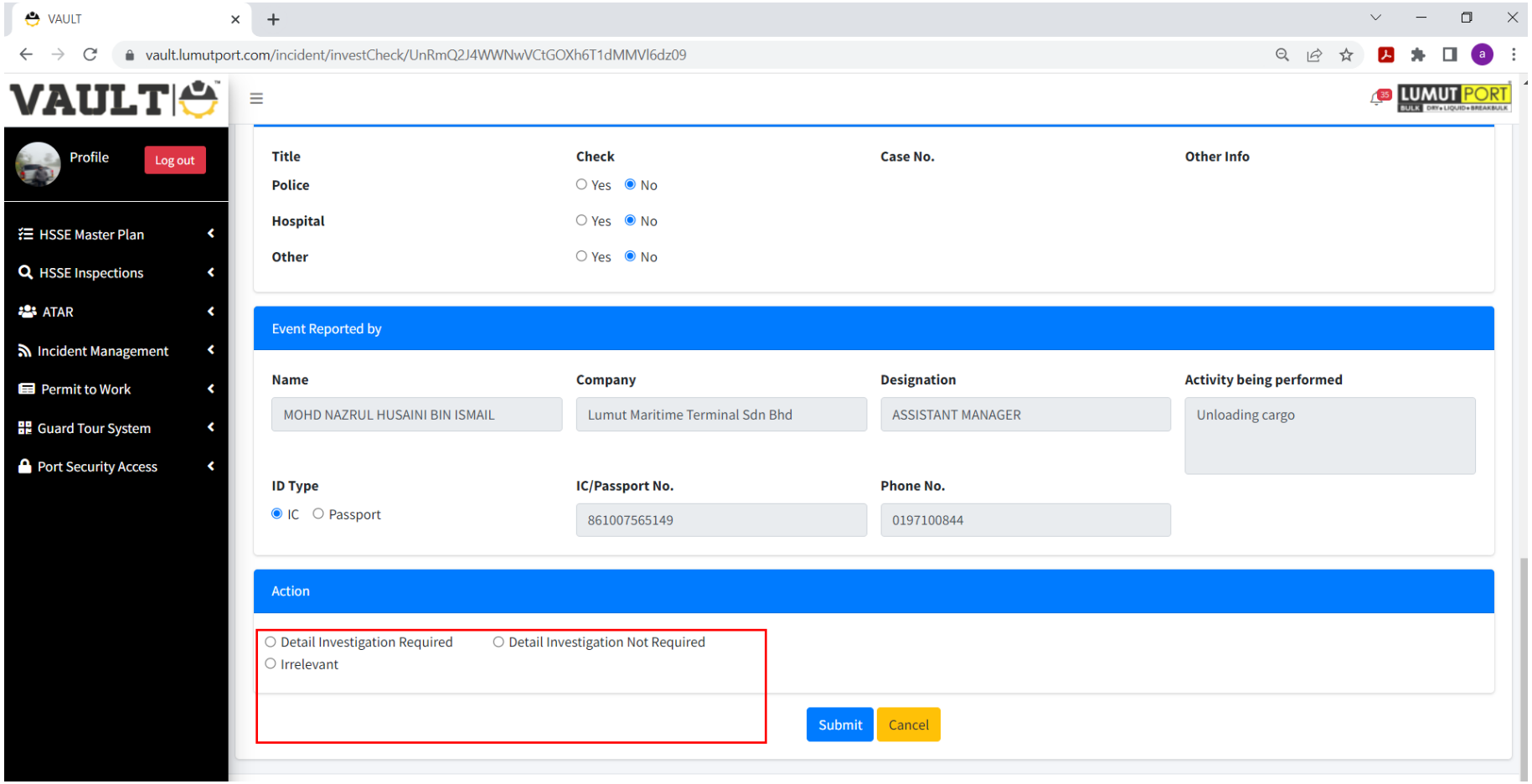
| Name | Date | Time |
|---|---|---|
| Bukhari Bin Daud | 04-05-2022 | 05:37 am |

# ITEM 2
## INCIDENT MANAGEMENT MODULE



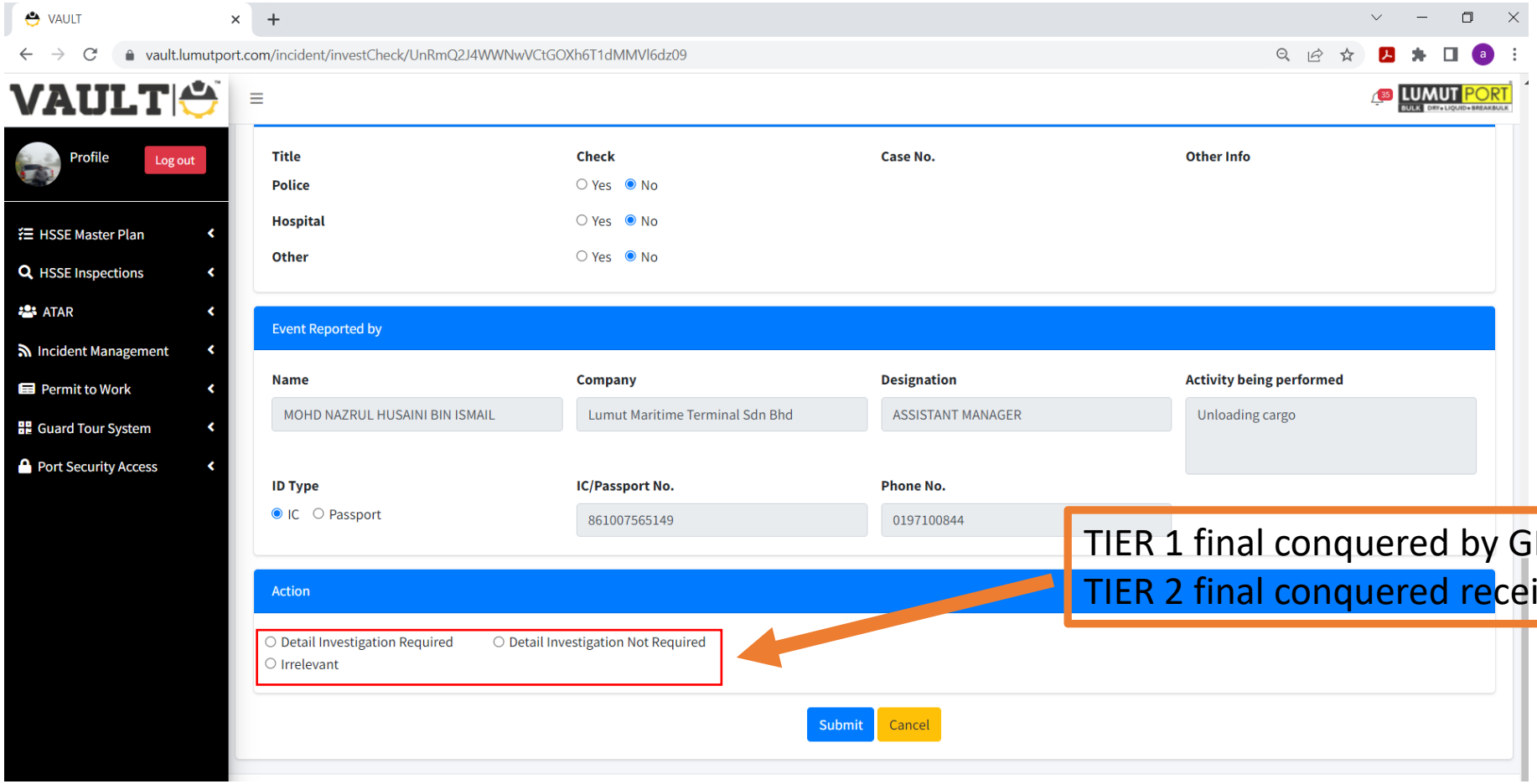
# ITEM 2.1 INCIDENT MANAGEMENT MODULE

- Module: INCIDENT MANAGEMENT
- Problem/Issue to Improve: To remove “Irrelevant” and “Detail-Investigation Not Required”. Changed to “Tier 1” and “Tier 2”



# ITEM 2.2 INCIDENT MANAGEMENT MODULE

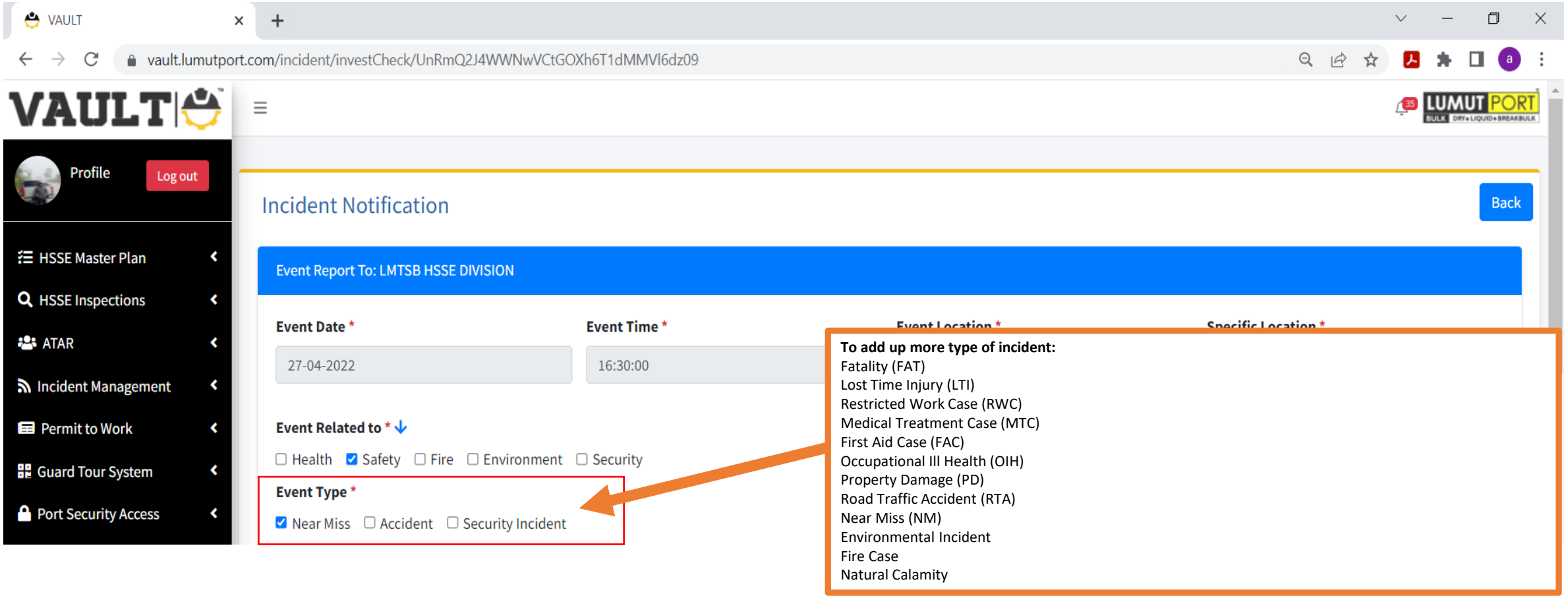
- Module: INCIDENT MANAGEMENT
- Problem/Issue to Improve: To add Tier 1 form (To be provided). For Tier 2 will utilized current form.
- For Tier 1 will be final conquered by GM Terminal/Verifier. For Tier 2, final conquered received by CEO/Approver.




TIER 1 final conquered by GM Terminal/Verifier.
TIER 2 final conquered received by CEO/Approver.

# ITEM 2.3 INCIDENT MANAGEMENT MODULE

- Module: INCIDENT MANAGEMENT
- Problem/Issue to Improve: To add Tier 1 form (To be provided). For Tier 2 will utilized current form.
- For Tier 1 will be final conquered by GM Terminal/Verifier. For Tier 2, final conquered received by CEO/Approver.



# ITEM 2.4 INCIDENT MANAGEMENT MODULE

- Module: INCIDENT MANAGEMENT
- Problem/Issue to Improve: To change “Verified by” to “Verified & Accepted by” at Verifier. To change “Approved by” to “Endorsed & Conquered by”

**Verified by**

| | |
|---|---|
| **Job Position** | GENERAL MANAGER LUMUT TERMINAL |
| **Name** | SYAMSYUL HELMY BIN MOHD RAFIAEE |
| **Verified Date & Time** | 2021-07-13 19:00:17 |
| **Verified Remark** | The recommended action completed. The repair cost has been paid by the ship agent. |

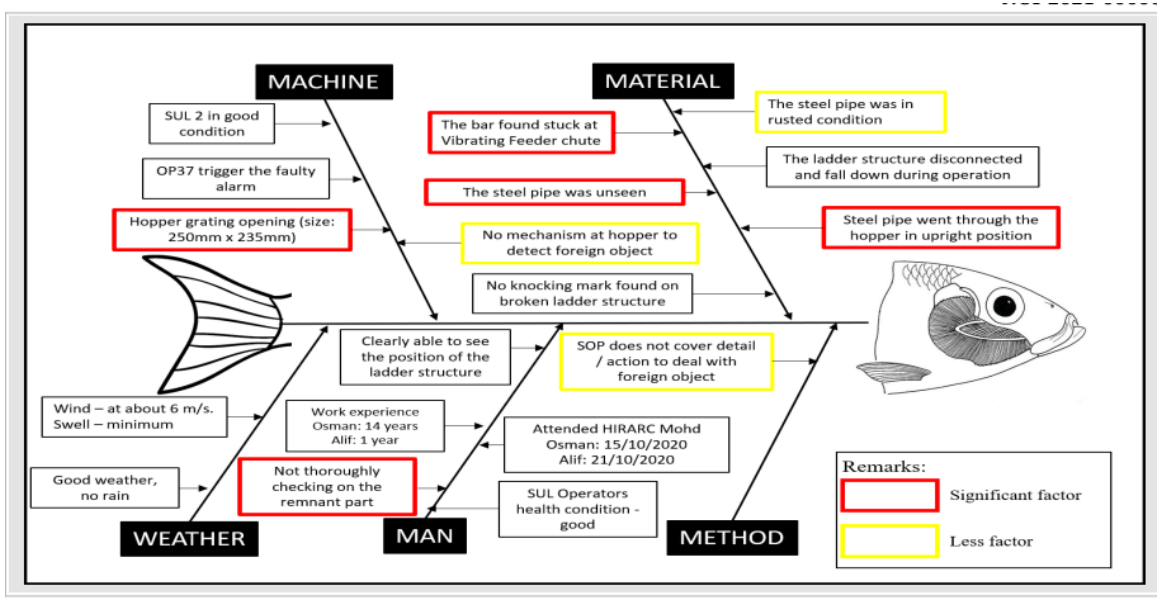
**Approved by**

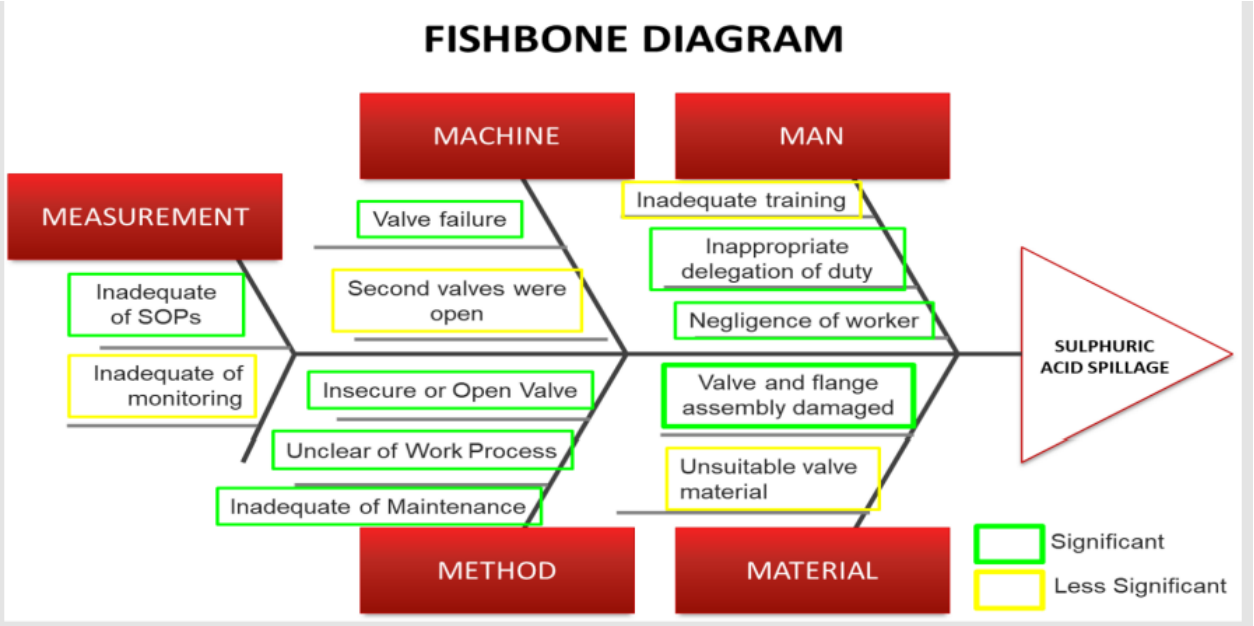
| | |
|---|---|
| **JobPosition** | CHIEF EXECUTIVE OFFICER |
| **Name** | MUBARAK ALI BIN GULAM RASUL |
| **Approved Date & Time** | 2021-07-15 13:34:16 |
| **Approved Remark** | 1. More effective measure need to be taken to ensure that this incident do not occur. 2. Do a trial run before the actual operations. Maybe at anchorage area or at wharf. 3. Preferably use own machines RHL MHL for loading operation instead of vessel. |

**“Verified by” to “Verified & Accepted by” at Verifier.**
**“Approved by” to “Endorsed & Conquered by”**

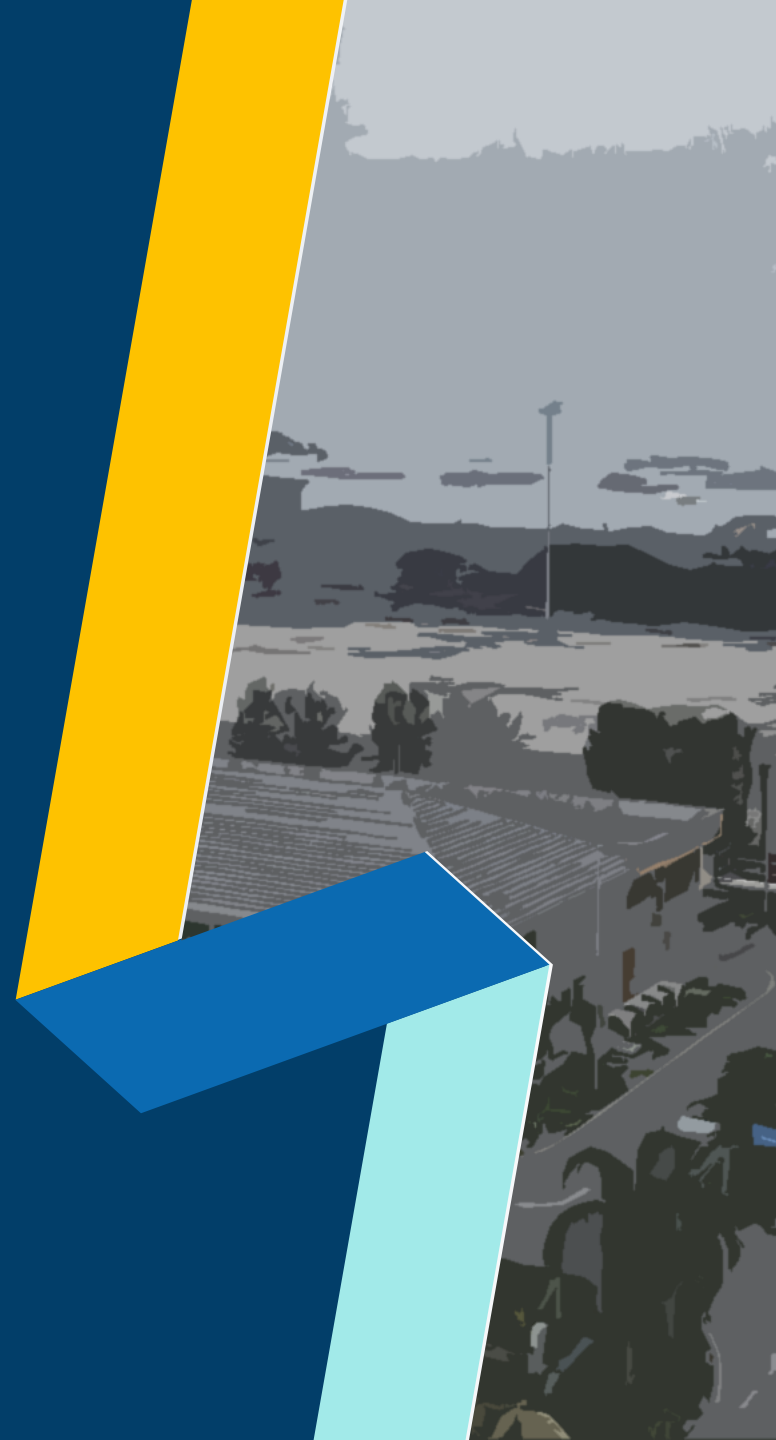
# ITEM 2.4 INCIDENT MANAGEMENT MODULE

- Module: INCIDENT MANAGEMENT
- Problem/Issue to Improve: To add template of Root Cause Analysis (RCA) diagram include investigation report. The sample of RCA will be provided.



# ITEM 3
## MONTHLY INSPECTION



# ITEM 3.1 MONTHLY INSPECTION MODULE

- Module: MONTHLY INSPECTION
- Problem/Issue to Improve: To change “Verified by” to “Verified & Accepted by” at Verifier. To remove Approver at flowchart of inspection.

| | | | | | |
|---|---|---|---|---|---|
| **INSPECTION ID** | : | MON-INSP-00018 | **INSPECTION DATE** | : | 24-03-2022 |
| **TERMINAL** | : | Lumut Terminal | **LOCATION** | : | OSR,SSD LMT Workshop |
| **INSPECTED BY** | : | ABDUL HANNAN BIN ENSAN | **REVIEWED BY** | : | MOHD RUSLY BIN TAMSIR |
| **VERIFIED BY** | : | MOHD RUSLY BIN TAMSIR | **APPROVED BY** | : | MUBARAK ALI BIN GULAM RASUL |

“Verified by” to “Verified & Accepted by” at Verifier.

To remove Approver at flowchart of inspection.

# ITEM 3.2 MONTHLY INSPECTION MODULE

- Module: MONTHLY INSPECTION
- Problem/Issue to Improve: All item title had mismatch. To repair and fix this issue according to checklist inserted in master. Including the pdf format.

| S/No | Visual Inspection |
|---|---|
| 1 | Work areas are clean and free from scattered materials and rubbish. |
| 2 | Walkways and passageways are clear. |
| 3 | Materials and equipment are stored properly. |
| 4 | Electrical cords, hoses and welding leads, etc. are elevated to prevent electrocution/trip hazards. |

| S/No | Functional Test | MAX SCORE | GIVEN SCORE | COMMENTS |
|---|---|---|---|---|
| 1 | Hard hats, safety boot, high visibility vest. | 3 | 3 | |
| 2 | Awareness/campaigns for proper use of PPEs. | 3 | 3 | |
| 3 | Damaged PPEs are not in use as required and replaced with new PPEs | 3 | 3 | |

| S/No | Automatic Safety Load Indicator | MAX SCORE | GIVEN SCORE | COMMENTS |
|---|---|---|---|---|
| 1 | Built per standard and specification. | 3 | 3 | |
| 2 | Proper/adequate access and egress provided | 3 | 3 | |
| 3 | Free from defects i.e. cracks, bends, nails etc. | 3 | 3 | |
| 4 | Tagged and inspected correctly. | 3 | 0 | not sign scaffolding tagging |
| 5 | Proper ladder for the job | 3 | 3 | |
| 6 | Ladder secured at the base and/or top. | 3 | 3 | |
| 7 | Ladder placed at proper angle and proper landing. | 3 | 3 | |
| 8 | Boards/planks are in safe condition | 3 | 3 | |
| 9 | Qualified inspector available. | 3 | 3 | |
| 10 | Damaged / Defective scaffold materials shall be segregated and marked with Red Do not use. | NA | NA | |
| 11 | Platforms free of debris and access clear of any obstruction. | 3 | 3 | |
| 12 | Scaffolding design for its purpose - Light, medium, heavy duty | 3 | 3 | |

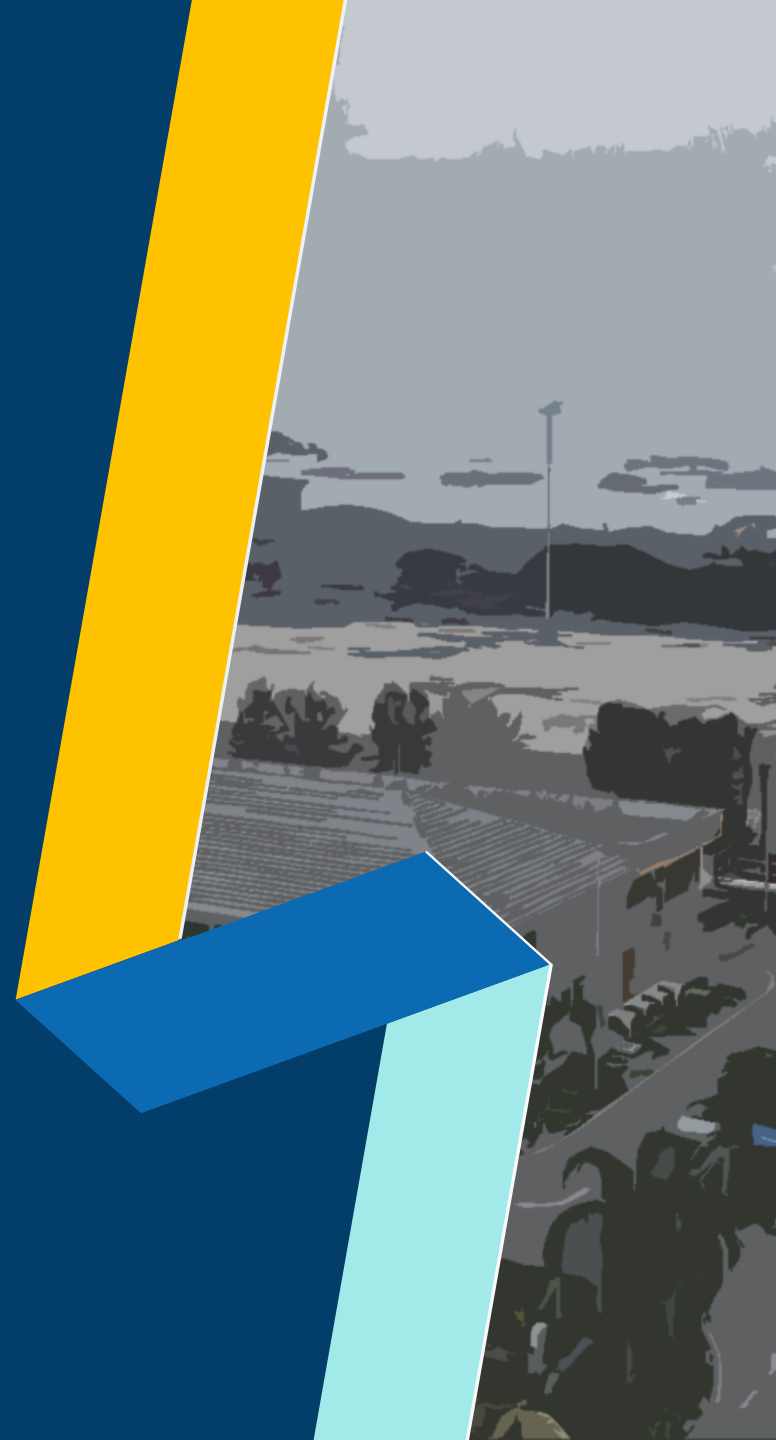
| | MAX SCORE | GIVEN SCORE | COMMENTS |
|---|---|---|---|
| | NA | NA | |
| e installed | NA | NA | |
| | NA | NA | |
| | NA | NA | |
| | NA | NA | |
| istance away from the edge | NA | NA | |
| ıer. | NA | NA | |
| | **0** | **0** | |

| S/No | Over Load Alarm | MAX SCORE | GIVEN SCORE | COMMENTS |
|---|---|---|---|---|
| 1 | Lights, brakes, horns and alarms working properly. | NA | NA | |
| 2 | Back up alarm audible enough to alert personnel around the vicinity. | NA | NA | |
| 3 | Inspections record is available. | NA | NA | |
| 4 | Trained and certified Banks men are available for moving equipment | NA | NA | |
| 5 | All vehicles and equipment free from any leaks and other defects | NA | NA | |
| 6 | Vehicles and equipment following site maximum speed limit | NA | NA | |
| 7 | Operator/driver cabin free of potential obstruction i.e. Curtain, card boards, etc. | NA | NA | |
| 8 | Adequate fire extinguishers are available and inspected/at machinery | NA | NA | |
| 9 | All licenses, certificates and other required | NA | NA | |
| 10 | Operator/driver cabin free of grease and debris. | NA | NA | |
| | **Total Points awarded:** | **0** | **0** | |

| S/No | Travelling Alarm | MAX SCORE | GIVEN SCORE | COMMENTS |
|---|---|---|---|---|
| 1 | Work station design i.e. tool selection, tool design | 3 | 3 | |
| 2 | Personnel posture vs tools and/or equipment - bending , twisting , etc | 3 | 3 | |
| 3 | Exposure to vibration | 3 | 3 | |
| 4 | Work platforms and stairs not too high not too low | 3 | 3 | |
| 5 | Floor surfaces - wet, slippery, smooth, vibration, etc | 3 | 3 | |
| 6 | Persons not lifting more than 25Kgs/person | 3 | 3 | |
| 7 | Persons getting enough work rest breaks | 3 | 3 | |
| | **Total Points awarded:** | **21** | **21** | |

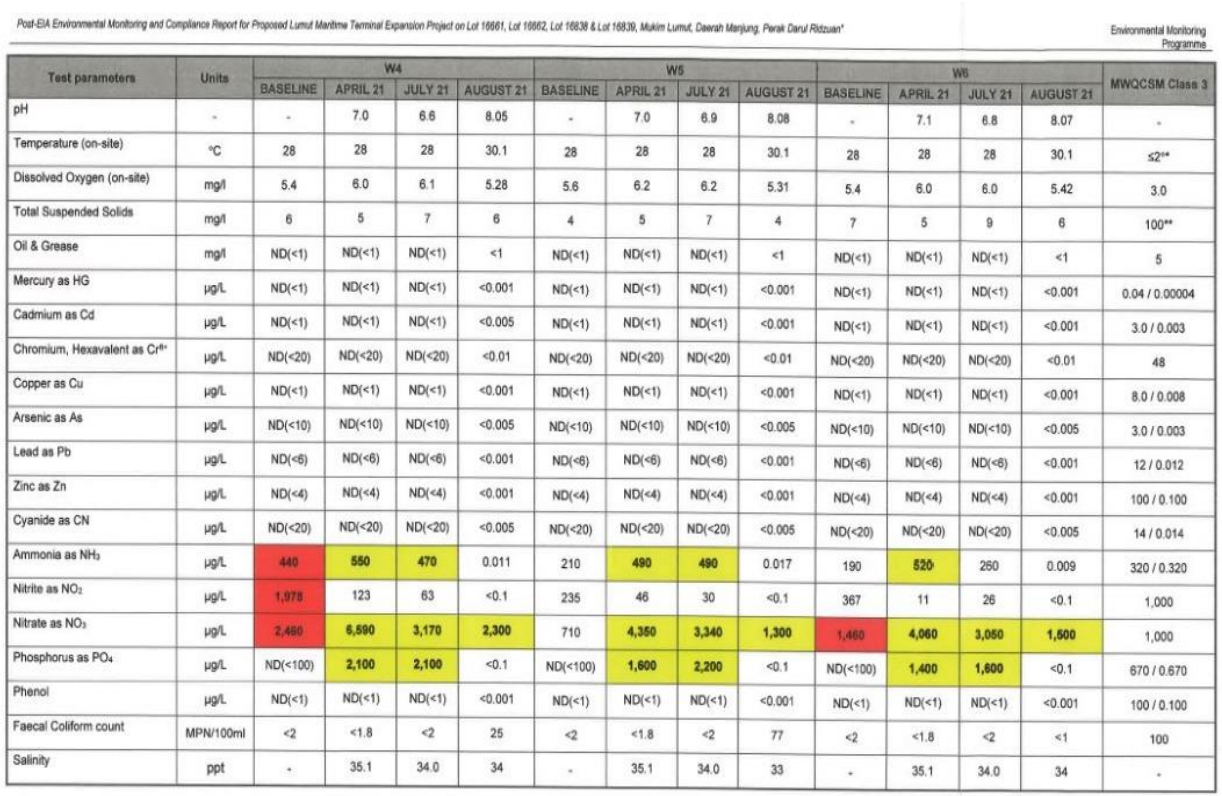
# ITEM 4

## ADDITIONAL MODULE FOR ENVIRONMENTAL MONITORING



# ITEM 4.1 ADDITIONAL MODULE FOR ENVIRONMENTAL MONITORING

- Module: NEW MODULE – Environmental Monitoring
- New Module to tracking and record keeping of Environmental Monitoring Report.



*Post-EIA Environmental Monitoring and Compliance Report for Proposed Lumut Maritime Terminal Expansion Project on Lot 16661, Lot 16662, Lot 16838 & Lot 16839, Mukim Lumut, Daerah Manjung, Perak Darul Ridzuan*

Environmental Monitoring Programme

| Test parameters | Units | W4 | | | | W5 | | | | W6 | | | | MWQCSM Class 3 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | BASELINE | APRIL 21 | JULY 21 | AUGUST 21 | BASELINE | APRIL 21 | JULY 21 | AUGUST 21 | BASELINE | APRIL 21 | JULY 21 | AUGUST 21 | |
| pH | - | - | 7.0 | 6.6 | 8.05 | - | 7.0 | 6.9 | 8.08 | - | 7.1 | 6.8 | 8.07 | - |
| Temperature (on-site) | °C | 28 | 28 | 28 | 30.1 | 28 | 28 | 28 | 30.1 | 28 | 28 | 28 | 30.1 | ≤2** |
| Dissolved Oxygen (on-site) | mg/l | 5.4 | 6.0 | 6.1 | 5.28 | 5.6 | 6.2 | 6.2 | 5.31 | 5.4 | 6.0 | 6.0 | 5.42 | 3.0 |
| Total Suspended Solids | mg/l | 6 | 5 | 7 | 6 | 4 | 5 | 7 | 4 | 7 | 5 | 9 | 6 | 100** |
| Oil & Grease | mg/l | ND(<1) | ND(<1) | ND(<1) | <1 | ND(<1) | ND(<1) | ND(<1) | <1 | ND(<1) | ND(<1) | ND(<1) | <1 | 5 |
| Mercury as HG | µg/L | ND(<1) | ND(<1) | ND(<1) | <0.001 | ND(<1) | ND(<1) | ND(<1) | <0.001 | ND(<1) | ND(<1) | ND(<1) | <0.001 | 0.04 / 0.00004 |
| Cadmium as Cd | µg/L | ND(<1) | ND(<1) | ND(<1) | <0.005 | ND(<1) | ND(<1) | ND(<1) | <0.001 | ND(<1) | ND(<1) | ND(<1) | <0.001 | 3.0 / 0.003 |
| Chromium, Hexavalent as $Cr^{6+}$ | µg/L | ND(<20) | ND(<20) | ND(<20) | <0.01 | ND(<20) | ND(<20) | ND(<20) | <0.01 | ND(<20) | ND(<20) | ND(<20) | <0.01 | 48 |
| Copper as Cu | µg/L | ND(<1) | ND(<1) | ND(<1) | <0.001 | ND(<1) | ND(<1) | ND(<1) | <0.001 | ND(<1) | ND(<1) | ND(<1) | <0.001 | 8.0 / 0.008 |
| Arsenic as As | µg/L | ND(<10) | ND(<10) | ND(<10) | <0.005 | ND(<10) | ND(<10) | ND(<10) | <0.005 | ND(<10) | ND(<10) | ND(<10) | <0.005 | 3.0 / 0.003 |
| Lead as Pb | µg/L | ND(<6) | ND(<6) | ND(<6) | <0.001 | ND(<6) | ND(<6) | ND(<6) | <0.001 | ND(<6) | ND(<6) | ND(<6) | <0.001 | 12 / 0.012 |
| Zinc as Zn | µg/L | ND(<4) | ND(<4) | ND(<4) | <0.001 | ND(<4) | ND(<4) | ND(<4) | <0.001 | ND(<4) | ND(<4) | ND(<4) | <0.001 | 100 / 0.100 |
| Cyanide as CN | µg/L | ND(<20) | ND(<20) | ND(<20) | <0.005 | ND(<20) | ND(<20) | ND(<20) | <0.005 | ND(<20) | ND(<20) | ND(<20) | <0.005 | 14 / 0.014 |
| Ammonia as $NH_3$ | µg/L | 440 | 550 | 470 | 0.011 | 210 | 490 | 490 | 0.017 | 190 | 520 | 260 | 0.009 | 320 / 0.320 |
| Nitrite as $NO_2$ | µg/L | 1,978 | 123 | 63 | <0.1 | 235 | 46 | 30 | <0.1 | 367 | 11 | 26 | <0.1 | 1,000 |
| Nitrate as $NO_3$ | µg/L | 2,460 | 6,590 | 3,170 | 2,300 | 710 | 4,350 | 3,340 | 1,300 | 1,460 | 4,060 | 3,050 | 1,500 | 1,000 |
| Phosphorus as $PO_4$ | µg/L | ND(<100) | 2,100 | 2,100 | <0.1 | ND(<100) | 1,600 | 2,200 | <0.1 | ND(<100) | 1,400 | 1,600 | <0.1 | 670 / 0.670 |
| Phenol | µg/L | ND(<1) | ND(<1) | ND(<1) | <0.001 | ND(<1) | ND(<1) | ND(<1) | <0.001 | ND(<1) | ND(<1) | ND(<1) | <0.001 | 100 / 0.100 |
| Faecal Coliform count | MPN/100ml | <2 | <1.8 | <2 | 25 | <2 | <1.8 | <2 | 77 | <2 | <1.8 | <2 | <1 | 100 |
| Salinity | ppt | - | 35.1 | 34.0 | 34 | - | 35.1 | 34.0 | 33 | - | 35.1 | 34.0 | 34 | - |

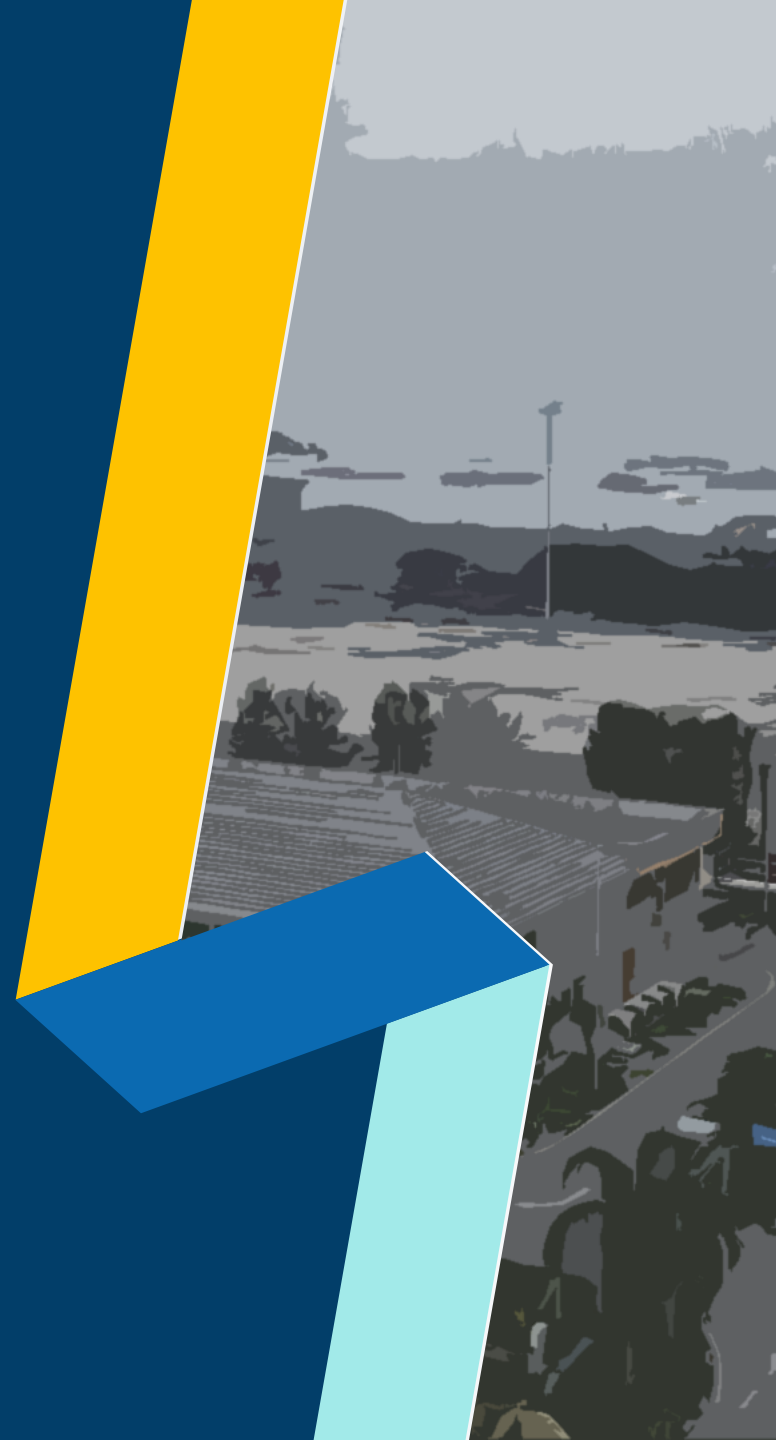
**Table 4.13: Environmental Noise Level Monitoring Results**

| Point | Date | Monitoring Time | Period | Total $L_{eq}$ dB(A) | Guidelines* |
|---|---|---|---|---|---|
| N1 | 16th August 2021 | 07:00 – 21:45 | Day Time | 67.8 | ≤ 55.0# |
| | 16th – 17th August 2021 | 22:00 – 06:45 | Night Time | 67.8 | ≤ 45.0# |
| N2 | 16th August 2021 | 07:04 – 21:49 | Day Time | 58.6 | ≤ 55.0# |
| | 16th – 17th August 2021 | 22:04 – 06:49 | Night Time | 47.7 | ≤ 45.0# |
| N3 | 16th August 2021 | 07:00 21:45 | | | |
| | 16th – 17th August 2021 | 22:00 06:45 | | | |

**Table 4.11: Ambient Air Quality Results**

| Parameter | Results | | | | | Air Quality Standard | |
|---|---|---|---|---|---|---|---|
| | A1 | A2 | A3 | A4 | A5 | Averaging Time | Standard 2020 |
| PM10 | 12 | 17 | 22 | 14 | 19 | 24 hours | 100 µg/m³ |
| Nitrogen Dioxide (NO2) | <0.5 | <0.5 | <0.5 | <0.5 | <0.5 | 24 hour | 70 µg/m³ |
| Sulphur Dioxide (SO2) | <5 | <5 | <5 | <5 | <5 | 24 hours | 80 µg/m³ |
| Carbon Monoxide (CO) | 1 | 0.7 | 0.6 | 0.7 | 0.7 | 8 hour | 10 mg/m³ |
| Ozone (O3) | <5 | <5 | <5 | <5 | <5 | 8 hour | 100 µg/m³ |

# ITEM 5
## GUARD TOUR SYSTEM (GTS)



# ITEM 5.1 GUARD TOUR SYSTEM (GTS)

- Module: GUARD TOUR SYSTEM (GTS)
- Problem/Issue to Improve:
- 1. Large storage required for photo
- 2. Propose new method such as GPS linked system at GTS Area.

